

Identifying the Key Role of Pyridinic-N–Co Bonding in Synergistic Electrocatalysis for Reversible ORR/OER

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For many regenerative electrochemical energy-conversion systems, hybrid electrocatalysts comprising transition metal (TM) oxides and heteroatom-doped (e.g., nitrogen-doped) carbonaceous materials are promising bifunctional oxygen reduction reaction/oxygen evolution reaction electrocatalysts, whose enhanced electrocatalytic activities are attributed to the synergistic effect originated from the TM–N–C active sites. However, it is still ambiguous which configuration of nitrogen dopants, either pyridinic or pyrrolic N, when bonded to the TM in oxides, predominately contributes to the synergistic effect. Herein, an innovative strategy based on laser irradiation is described to controllably tune the relative concentrations of pyridinic and pyrrolic nitrogen dopants in the hybrid catalyst, i.e., NiCo₂O₄ NPs/N-doped mesoporous graphene. Comparative studies reveal the dominant role of pyridinic-N–Co bonding, instead of pyrrolic-N bonding, in synergistically promoting reversible oxygen electrocatalysis. Moreover, density functional theory calculations provide deep insights into the corresponding synergistic mechanism. The optimized hybrid, NiCo/NLG-270, manifests outstanding reversible oxygen electrocatalytic activities, leading to an overpotential different ΔE among the lowest value for highly efficient bifunctional catalysts. In a practical reversible Zn–air battery, NiCo/NLG-270 exhibits superior charge/discharge performance and long-term durability compared to the noble metal electrocatalysts.

electrode reactions, including the oxygen reduction reaction (ORR) and the oxygen evolution reaction (OER), always play vital roles, however, suffering from intrinsic sluggish kinetics due to multiple electron transfer processes.^[2] At present, precious metal Pt- and Ru (or Ir)-based catalysts are known to exhibit the best electrochemical activities for catalyzing ORR and OER,^[3] respectively, however, their prohibitive cost, scarcity, and poor durability have severely retarded large-scale implementation of the above-mentioned renewable energy systems. In addition, it is of great challenges to rationally design and exploit efficient bifunctional ORR/OER electrocatalysts, necessary to the regenerative operation.^[4]

Numbers of recent studies have shown that hybrid electrocatalysts, composed of earth-abundant transition metal (TM) oxides and heteroatom-doped (e.g., nitrogen-doped) carbonaceous materials, usually exhibit exceptional bifunctional ORR/OER activities and good stability superior to their single-component counterparts.^[4a,5] The improved electro-

The booming economy relied on traditional fossil fuels has led to ever-increasing concerns about the global environment and energy challenges. It is, hence, highly demanded to exploit sustainable and green energy conversion and storage technologies, such as regenerative fuel cells, rechargeable metal–air batteries, and water electrolyzers.^[1] Among these devices, oxygen

chemical performance of the composites has been generally ascribed to the followings. (1) The strong chemical bonding between metal-containing species and N-doped-carbon supports, due to the high affinity of nitrogen (especially pyridinic and pyrrolic N) to the TM ions, can greatly facilitate the interfacial electron transfer for the hybrid electrocatalysts.^[4a,6]

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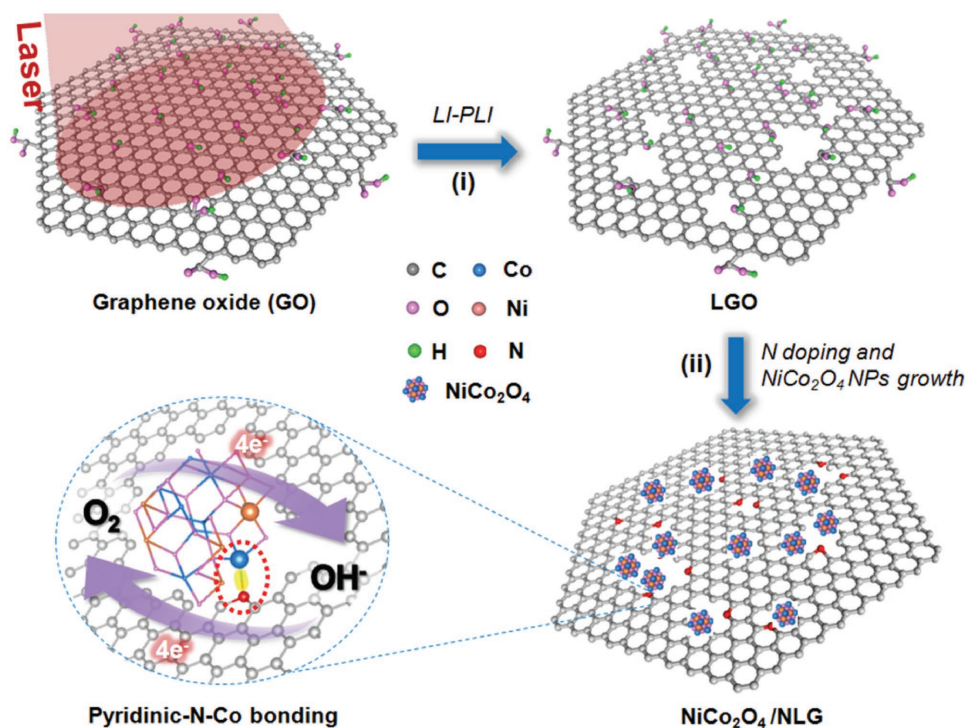
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(2) More importantly, the strong coupling between TM and N-carbon at the interface can normally induce a so-called synergistic effect for the hybrids,^[5a,7] benefiting substantially enhanced oxygen electrochemistry compared to their counterparts without nitrogen doping. Nevertheless, up to date, the fundamental mechanism of synergistic electrocatalysis for ORR/OER is still ambiguous. In particular, although it was evidenced that the M–N–C (M = Fe, Co, Cu, etc.) active sites played an important role in boosting the electrocatalytic activities of metal-oxide/N-carbon hybrids,^[4a,5a,c,8] little has been done to distinguish the contributions from M–N bonds with different nitrogen configurations. In other words, it is still unclear which configuration of nitrogen dopants, either pyridinic or pyrrolic N, when bonded to TM in oxides, predominately contributes to the synergistic effect, and how it would cooperate with the metal ions at the hybrid interface to synergistically catalyze both oxygen reduction and evolution processes. Recently, Yu et al. speculated that the pyridinic N coordinated with iron in $\gamma\text{-Fe}_2\text{O}_3/\text{N-doped carbon nanotubes}$ was the primary ORR active site.^[9] However, the contribution from pyrrolic N was not ruled out, moreover, for other catalytically active TM ions, e.g., Co or Ni which are normally active sites not only for ORR but also for OER, it is still unknown which configuration of nitrogen dopants will help to synergistically promote the reversible oxygen electrocatalysis for the hybrids. The research bottle-neck essentially lies in the fact that it is rather hard to selectively tune the concentration of nitrogen dopants with different configurations. It was reported recently that thermal treatment of a mixture of iron precursors and rGO under ammonia gas would tune the concentration of pyridinic N by varying the annealing time at 900 °C, whereas the high-temperature annealing unavoidably resulted

in large NPs (up to 100 nm) with low specific surface areas,^[10] degrading the catalytic activities.

Inspired by the motivation to exploit highly efficient bifunctional ORR/OER electrocatalysts via engineering the interfacial structure of composite catalysts, herein, an innovative strategy, based on low-intensity pulsed-laser irradiation (LI-PLI), is developed to controllably and selectively regulate the relative contents of pyridinic and pyrrolic N in $\text{NiCo}_2\text{O}_4/\text{N-graphene hybrids}$ ($\text{NiCo}_2\text{O}_4/\text{NLG}$, or NiCo/NLG for short). As shown in **Scheme 1**, the LI-PLI method is capable of tuning the in-plane mesopores density of the resulted mesoporous graphene oxides (LGO), thus in turn, facilitating engineering the relative concentrations of pyridinic and pyrrolic N during the subsequent hydrothermal treatments for nitrogen doping and small NiCo_2O_4 NPs growth. Strong chemical bonding between N dopants and Co ions were confirmed. The underlying correlation between different nitrogen configurations in Co–N–C bonds and the synergistically enhanced oxygen electrochemistry (ORR/OER) for the hybrid catalysts were experimentally investigated, and eventually, deep insights into the mechanism of the synergistic effect were acquired in combination of experiments and density functional theory (DFT) calculations. The pyridinic-N–Co bonding, instead of pyrrolic-N–Co bonding, was identified as the key for the synergistically promoted ORR and OER. The best composite ($\text{NiCo}/\text{NLG-270}$) exhibits outstanding reversible ORR/OER performance, surpassing most of the reported bi-functional electrocatalysts, as well as a combination of benchmark noble-metal catalysts ($\text{Pt}/\text{C} + \text{RuO}_2$). Furthermore, when applied as the air electrode for a rechargeable Zn–air battery, $\text{NiCo}/\text{NLG-270}$ demonstrates excellent charge/discharge behavior and long-term stability superior to the noble-metal based electrocatalysts.



Scheme 1. Schematic illustration of the preparation of $\text{NiCo}_2\text{O}_4/\text{NLG}$ hybrid for reversible oxygen electrocatalysis.

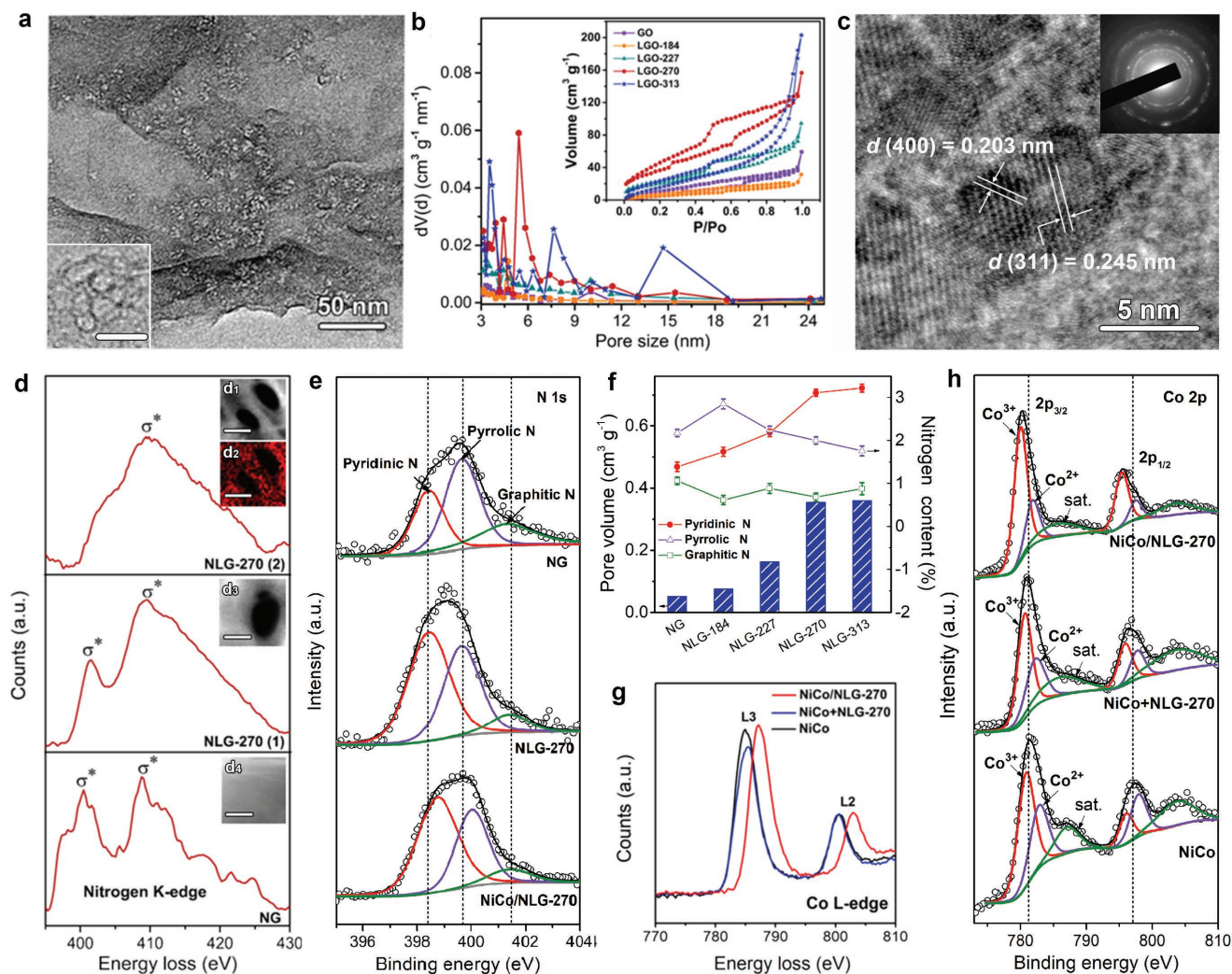


Figure 1. a) TEM and HR-TEM (inset) images of LGO-270, showing the in-plane porous structure. Scale bar in the inset represents 10 nm. b) Pore size distribution of GO and LGO samples. Inset illustrates the corresponding N_2 adsorption–desorption isotherm curves. c) HR-TEM image of $NiCo_2O_4$ nanoparticles anchored on a NLG-270 nanosheet. Inset displays the corresponding SAED pattern. d) Nitrogen K-edge EELS spectra of NG and NLG-270. Two regions of NLG-270 with different local pore densities were examined, namely, NLG-270 (1) and NLG-270 (2), respectively. Insets show the STEM images (d_1 , d_3 , d_4) and the N energy-filtered TEM image (d_2), where the scale bars represent 20 nm. e) High-resolution XPS N 1s spectra of NiCo/NLG-270, NLG-270, and NG. f) The variation of pyridinic, pyrrolic, and graphitic N contents with pore volume among different samples. g) Cobalt L-edge EELS spectra and h) high resolution XPS Co 2p spectra of NiCo/NLG-270, NiCo+NLG-270, and $NiCo_2O_4$ NPs.

Laser beam with different energies, i.e., 184, 227, 270, and 313 mJ, were adopted in LI-PLI (see details in the Supporting Information), and the as-obtained products were denoted as LGO- x ($x = 184, 227, 270, 313$). The mesoporous structure is observed by using transition electron microscope (TEM). Distinct to the smooth surface of GO (Figure S1a, Supporting Information), a large number of in-plane mesopores smaller than 20 nm are created by laser irradiation, randomly distributed over the LGO sheet (Figure 1a). Interestingly, as the laser energy increases, the pore density appears to be much higher for the resultant LGO sheets (Figure S1b–f, Supporting Information). Thorough examination on the porous structure was further performed by adopting nitrogen adsorption analyses. The adsorption isotherms of LGO resemble type IV with an obvious hysteresis loop (inset of Figure 1b), indicating the mesopores formation after laser treatments.^[11] The

corresponding pore size distribution also confirms the presence of mesopores with size less than 20 nm (Figure 1c), in good agreement with the TEM images (inset of Figure 1a). The mesopore size distribution barely changes with laser beam energy, however, the pore density apparently augments and the pore volume reaches a value for LGO up to five times that of the original GO sheets (Figure S2, Supporting Information). The mesoporous defects in LGO are further verified by the relatively higher I_D/I_G ratio (I_D and I_G represents the intensity of D- and G-bands of graphene, respectively) in the Raman spectral of LGO-270 with respect to GO (0.94 vs 0.88, see Figure S3 in the Supporting Information).^[12] These results suggest that LI-PLI is an efficient approach to introduce considerable amount of in-plane mesopores onto GO sheets and also capable of engineering the pore density. Furthermore, the Fourier transform infrared spectroscopy (FTIR) measurements show that the

surface oxygen functionalities ($-\text{OH}$, $-\text{OOH}$, etc.) are retained in LGO (Figure S4, Supporting Information), which is also confirmed by the X-ray diffraction (XRD) patterns (Figure S5, Supporting Information), where the diffraction peak around 10.2° corresponds to the (001) reflection of oxidized graphite present in both GO and LGO.^[13]

The engineering of the relative contents of pyridinic and pyrrolic nitrogen was achieved by doping nitrogen atoms into GO and LGO- x with different mesopore densities via a conventional hydrothermal route using ammonia as nitrogen precursor (see details in the Supporting Information), and the doped samples were named as NG and NLG- x ($x = 184, 227, 270, 313$), respectively. X-ray photoelectron spectroscopy (XPS) survey data show that nitrogen dopants are effectively incorporated into the graphene sheets (Figure S6, Supporting Information). Both FTIR spectra (Figure S4, Supporting Information) and XRD patterns (Figure S5, Supporting Information) suggest that the N-doped samples are effectively reduced. The total nitrogen doping level slightly increases for samples with higher mesopore densities (Figure S7 and Table S1, Supporting Information). The scanning transmission electron microscopy (STEM) and N energy-filtered TEM images (insets d_1 , d_2 of Figure 1d) exhibit a homogeneous distribution of nitrogen atoms near the mesopore structure. In addition, electron energy loss spectroscopy (EELS) spectra of N K-edge (Figure 1d) were taken to get detailed information on nitrogen configurations. All the spectra are normalized based on the intensity of the peak at 408.4 eV. For NG with little in-plane pores (inset d_4 of Figure 1d,f), two prominent peaks are observed, where the one around 401 eV includes contributions from both pyrrolic N (400.9 eV) and graphitic N (401.9 eV), and the one located at 408.4 eV is attributed to the pyridinic N.^[14] In contrast, in the vicinity of mesoporous structures of NLG-270 with a much higher pore density (inset d_3 of Figure 1d,f), the peak intensity of pyridinic N promotes remarkably relative to that of both pyrrolic and graphitic N (Figure 1d), compared to NG. Besides, the relative peak intensity of pyridinic N further elevates in the region with a higher localized pore density (inset d_1 of Figure 1d). These results indicate that the laser-irradiation-induced mesopore defects with more edges preferentially facilitate the doping of pyridinic N instead of pyrrolic and graphitic N. The carbon K-edge EELS spectra (Figure S8, Supporting Information) prove that the laser treatment do not alter the sp^2 hybridization state of graphene.^[15] Additionally, high-resolution XPS measurements were carried out to establish quantitative relationship between the content of different nitrogen configurations and the mesopore densities. The N 1s spectra of both NG and NLG- x could be deconvoluted into pyridinic N (≈ 398.4 eV), pyrrolic N (≈ 399.6 eV), and graphitic N (≈ 401.4 eV) (Figure 1e and Figure S9 in the Supporting Information).^[14a] Meanwhile, the concentration of pyridinic N increases monotonically from 1.4% to 3.2% as the mesopore density gets higher (Figure 1f and Table S1 in the Supporting Information), whereas, the concentration of pyrrolic N reduces from 2.2% to 1.8% and the content of graphitic N hardly varies ($\approx 1.0\%$). This is in good agreement with the EELS data and again testifies that, by engineering the mesopore density in the graphene sheets via LI-PLI, one could feasibly and selectively tune the concentration of pyridinic N relative to that of pyrrolic or graphitic N. Thus, a further growth of spinel (NiCo_2O_4 , or

NiCo for short) NPs onto NLG- x layers would allow us to quantitatively probe the correlation between nitrogen configurations in N-metal bonds and the synergistically promoted oxygen electrocatalysis in the hybrid catalysts.

For the composite catalysts, same amounts of NiCo_2O_4 NPs were grown on NG and NLG- x that possess different relative contents of pyridinic and pyrrolic N. The corresponding XRD pattern (Figure S10a, Supporting Information) matches well with the spinel structure (PDF No. 20-0781). Both Co 2p and Ni 2p XPS spectra of the hybrid can be deconvoluted into Co^{3+} , Co^{2+} , Ni^{3+} , Ni^{2+} , and their shakeup satellites (denoted as “sat.” for short, Figure 1h; Figure S11a, Supporting Information). The mixed valence states of Co and Ni ions strongly evidence that the NiCo_2O_4 NPs have an “inverse” spinel structure, where both Co^{3+} and Co^{2+} occupy tetrahedral sites and Ni^{3+} , Ni^{2+} , and Co^{3+} occupy octahedral sites (Figure S12, Supporting Information).^[16] The NPs anchored on the graphene sheets are uniform in size (≈ 7.1 nm) (Figure S10b and inset, Supporting Information). The high-resolution TEM (HR-TEM) implies that the NPs have high crystalline with lattice spacing of 0.203 and 0.245 nm, expected for (400) and (311) crystal planes for NiCo_2O_4 lattice (Figure 1c). Besides, the diffraction rings in the selected area electron diffraction (SAED) pattern further verifies the spinel structure of the NPs (inset of Figure 1c). The elemental mapping of the composite catalyst suggests that Ni and Co are dominantly distributed in the position of NPs, while C, N, and O are uniformly dispersed all over the sample (Figure S10c, Supporting Information). Among all the hybrid samples, the weight percentage of NiCo_2O_4 are kept to be the same, around 51 wt% (Table S2, Supporting Information) as determined by the inductively coupled plasma mass spectroscopy. The strong nitrogen-metal chemical bonding for the hybrids were confirmed by high-resolution XPS and EELS analyses. For the composite catalyst (NiCo/NLG-270), an obvious shift (≈ 0.3 – 0.4 eV) of pyridinic and pyrrolic N 1s XPS peaks toward higher binding energy compared to those for NLG-270 was observed (Figure 1e), implying that the pyridinic and pyrrolic N are probably coordinated with Co or Ni ions at the hybrid interface.^[9,10] However, the fixed peak position of graphitic N 1s XPS spectra implies no chemical bonds forming between the graphitic N and metal ions. On the other hand, in the high resolution Co 2p scan, both Co $2\text{p}_{3/2}$ and Co $2\text{p}_{1/2}$ peaks for NiCo/NLG-270 shift to lower binding energy, corresponding to a higher oxidation state of Co ions (Figure 1h), compared to the isolated NiCo_2O_4 NPs and NiCo+NLG-270 (a physical mixture of NiCo_2O_4 NPs and NLG-270), suggesting strong chemical bond forming between Co and pyridinic and/or pyrrolic N with a high electronegativity, leading to partial electron migration from Co to pyridinic and/or pyrrolic N.^[14a] Meanwhile, the Co L-edge EELS spectrum of NiCo/NLG-270 also shift to higher energy loss range relative to NiCo_2O_4 NPs and NiCo+NLG-270 (Figure 1g), which can be ascribed to Co coordinated with pyridinic and/or pyrrolic N, in good agreement with the XPS results. In contrast, no discernible peak shift was observed in both Ni 2p XPS and Ni L-edge EELS spectra for NiCo/NLG-270 (Figure S11, Supporting Information), implying that Ni ions seldom bond with the nitrogen atoms. Additionally, XPS tests (Figure 1e and Figure S13a–e, Supporting Information) were adopted to examine the relative contents of nitrogen with different

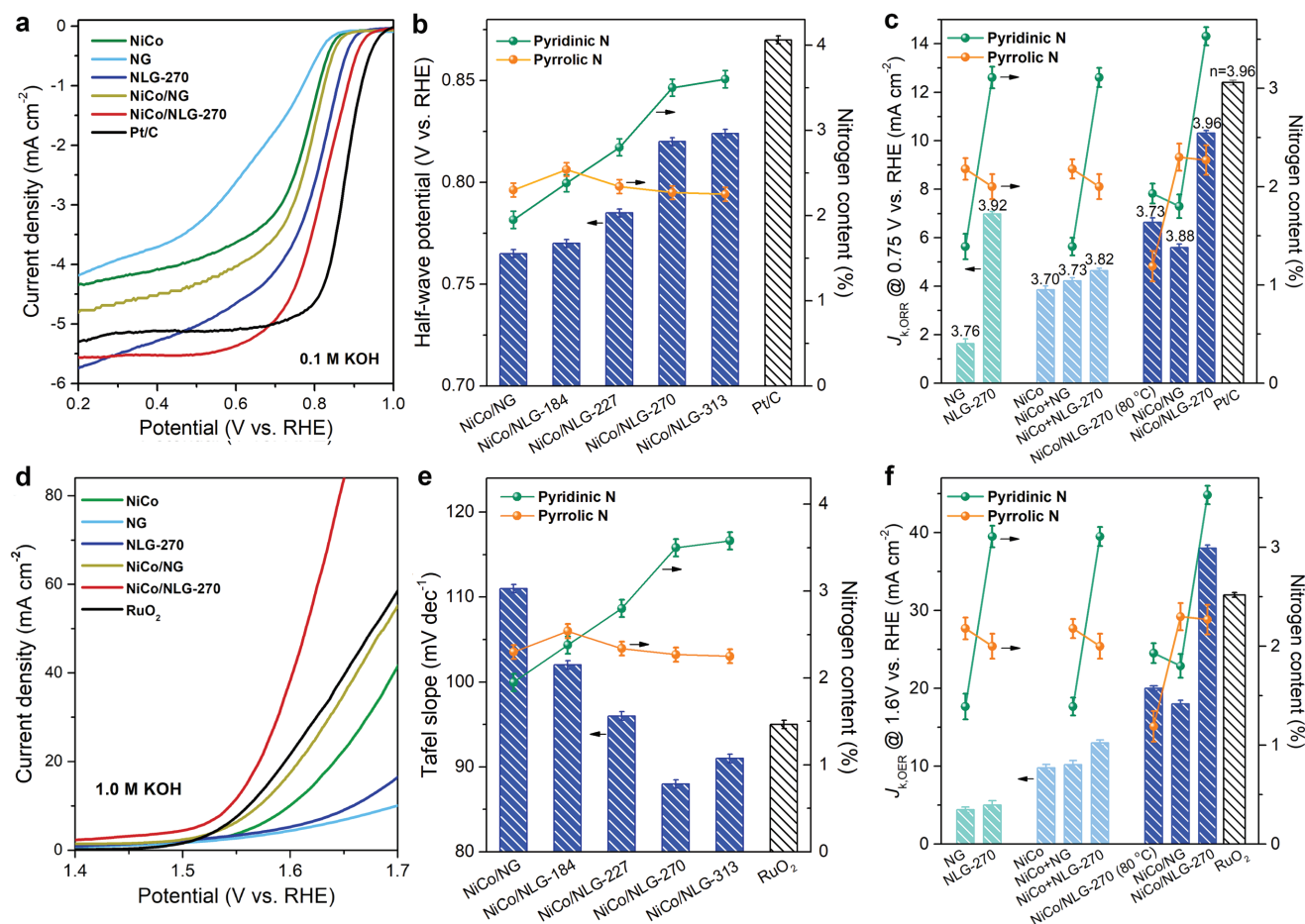


Figure 2. a) ORR LSV curves of NG, NLG-270, NiCo₂O₄ NPs, NiCo/NG, NiCo/NLG-270, and Pt/C, in 0.1 M KOH solution (rotation rate: 1600 rpm). b) ORR half-wave potential of the hybrid catalysts with different contents of pyridinic and pyrrolic N bonded with Co ions, and Pt/C. c) Comparison of the kinetic currents and the electron transfer number for the ORR process of the control samples, along with the corresponding contents of pyridinic and pyrrolic N bonded with Co. d) ORR LSVs of NG, NLG-270, NiCo₂O₄ NPs, NiCo/NG, NiCo/NLG-270, and Pt/C, in 1.0 M KOH solution. e) OER Tafel plots of the hybrid catalysts with different contents of pyridinic and pyrrolic N bonded with Co ions, and RuO₂. f) Comparison of the kinetic currents for the OER process of the control samples, along with the corresponding contents of pyridinic and pyrrolic N bonded with Co.

configurations for the hybrids. As shown in Figure S13f and Table S3 (Supporting Information), the concentration of pyridinic N increases monotonically from 1.8% to 3.5%, while the pyrrolic N and graphitic N hardly change, for the composite catalysts.

We further explore the correlation between nitrogen configuration in N–Co bonds and the reversible oxygen electrocatalysis performance of the hybrid catalyst NiCo/NLG-*x*. First of all, as expected, the linear sweep voltammograms (LSV) show that all the composites exhibit better ORR and OER catalytic activities than their constituent species and the physically mixed samples (Figure 2a,d and Figure S14, Supporting Information), which is normally attributed to the synergistic coupling effect between N-doped graphene and the metal oxides NPs.^[5a,b,17] Meanwhile, NiCo/NLG-270 and NiCo/NG also possess enhanced ORR/OER performance compared to their counterparts without nitrogen doping, i.e., NiCo/LG-270 and NiCo/G (Figure S15, Supporting Information), verifying again the synergistic effect of Co–N–C species. Most importantly, herein, we observe an additional promotion of both ORR and OER intrinsic activities primarily

correlated to the increasing concentration of pyridinic-N–Co bonds, rather than pyrrolic-N–Co bonds, thus providing experimental evidence toward the insights into the synergistic effect, as will be elaborated below. Specifically, for the ORR process, the onset potential remarkably move to more positive range by up to ≈100 mV for hybrid catalysts with a growing content of pyridinic-N–Co bonds, and achieve 0.96 V for NiCo/NLG-270, meanwhile, the diffusion-limited current density also substantially goes up and finally exceeds that of the benchmark 20 wt% Pt/C (Figure 2a and Figure S16a, Supporting Information). The half-wave potential also shifts cathodically, by up to ≈65 mV, with increasing pyridinic-N contents (Figure 2b). Therefore, it is highly possible that the pyridinic-N–Co bonds dominantly contribute to the ORR activity improvement. In contrast, for the control samples, i.e., the physical mixtures NiCo+ NLG-*x* (NG), which possess similar N doping level and mesoporous structures to the corresponding hybrids, but few N–Co bonds (Figure 1g,h), the LSV curves in the kinetic-controlled region are seldom affected by both pyridinic N contents and mesopore densities, leading to unchanged onset potential and half-wave

potential (Figure S16b, Supporting Information). This observation verifies again that the pyridinic-N-Co bonds play an important role in the promotion of oxygen reduction. Next, we quantitatively analyze contributions from pyridinic- and pyrrolic-N-Co bonds to the ORR intrinsic activity by analyzing the electrochemical kinetic parameters based on the Koutechy-Levich (K-L) equation. The kinetic current density ($J_{K, ORR}$) with mass-transport correction was derived from the rotating disk electrode data (Figures S17–S20, Supporting Information), and the apparent number of electrons transferred (n) was obtained by using the rotating ring-disk electrode data (Figure S21, Supporting Information). In Figure 2c, from the intrinsic activity perspective, we illustrate emphatically the synergistic effect for ORR and its positive correlation to the pyridinic-N-Co bonds in the composite catalysts. First of all, for N doped graphene itself (NG and NLG-270), $J_{K, ORR}$ increases in proportion to the pyridinic-N content, suggesting that pyridinic N is likely responsible to catalyzing the ORR process in N-graphene, consistent with previous findings in literatures.^[18] Besides, the higher porosity of NLG-270 might also contribute to the improved ORR activity. Afterward, when simply mixed with $NiCo_2O_4$ NPs, $J_{K, ORR}$ is reduced for NiCo+NLG-270, probably due to the coverage of active sites on NLG-270 by the less active $NiCo_2O_4$ NPs, which obviously has lower $J_{K, ORR}$ than NLG-270. A comparison of NiCo+NG and NiCo+NLG-270 shows that if without N-Co bonds, the increasing of pyridinic-N content and porosity could not effectively enhance $J_{K, ORR}$. Contrarily, in the case of hybrid catalysts, NiCo/NG and NiCo/NLG-270, where pyridinic or pyrrolic N has proved to be strongly coordinated with Co ions (Figure 1e,g,h), $J_{K, ORR}$ is dramatically promoted compared to the mixtures NiCo+NG and NiCo+NLG-270 without N-Co coordination. This implies that the formation of pyridinic- or pyrrolic-N-Co bonds induces strong synergistic effect. More importantly, $J_{K, ORR}$ for the hybrid catalysts increases with the concentration of pyridinic N bonded with Co, and $J_{K, ORR}$ of NiCo/NLG-270 is nearly twice as large as that of NiCo/NG, even approaching that of Pt/C. BET measurements show that the specific surface area (SSA) of NiCo/NLG-270 ($159.6 \text{ m}^2 \text{ g}^{-1}$) is lower than that of NiCo/NG ($330.3 \text{ m}^2 \text{ g}^{-1}$) (Figure S22a, Supporting Information). To eliminate the SSA effect, $J_{K, ORR}$ was normalized with respect to the corresponding surface area. After the normalization, NiCo/NLG-270 exhibits nearly three-fold enhancement in ORR activity compared to NiCo/NG (Figure S22b, Supporting Information). Moreover, hybrid samples with different pyrrolic-N contents and almost fixed pyridinic-N contents were also prepared (Figure S23a,b, Supporting Information), however, in this case, $J_{K, ORR}$ still follows the trend of pyridinic-N content, while no such correlation was found for pyrrolic N (Figure 2c and Figure S23c, Supporting Information). These observations indicate that pyridinic-N-Co bonds are dominantly responsible to the synergistic coupling effect substantially boosting the intrinsic ORR activity for the spinel/N-graphene hybrids. Meanwhile, the increasing contents of pyridinic-N bonded with Co further enhance the four-electron selectivity in the ORR process, verifying again the key role of pyridinic-N-Co bonds in facilitating electrocatalytic oxygen reduction.

On the other hand, the pyridinic-N-Co bonds also greatly contribute to the improvement of OER activity. As the pyridinic-N-Co bonds contents increases, the overpotential at

current density of 10 mA cm^{-2} reduces monotonically, by up to $\approx 40 \text{ mV}$ (Figure 2d and Figure S24a, Supporting Information), and finally becomes much lower than that of the benchmark OER catalyst RuO_2 . Meantime, the Tafel slope also remarkably decreases with increasing concentration of pyridinic-N-Co bonds, with the lowest Tafel slope smaller than that of RuO_2 for the best catalyst NiCo/NLG-270 (Figure 2e), suggesting that the pyridinic-N-Co bonds is capable of accelerating the kinetics of electrocatalytic oxygen evolution. In contrast, for mixtures NiCo+NG and NiCo+NLG-270 without N-Co bonds, the variation of pyridinic N contents and mesopore structures have little effects on both overpotential and Tafel slope (Figure S24b, Supporting Information), verifying again the key role of pyridinic-N-Co bonds in facilitating the OER process. We also quantitatively study the correlation between the intrinsic OER activity and the pyridinic- or pyrrolic-N-Co bonds by calculating the kinetic OER current ($J_{K, OER}$) at an overpotential of 0.37 V . For plain N-doped graphene (NG and NLG-270), an increasing of pyridinic-N content does not cause much growth of $J_{K, OER}$ (Figure 2f), indicating that the pyridinic N in N-graphene could not promote OER. When physically mixing NG or NLG-270 with $NiCo_2O_4$ NPs, $J_{K, OER}$ for NiCo+NG or NiCo+NLG-270 becomes more than twice that for NG or NLG-270, respectively, mainly due to the active sites on $NiCo_2O_4$ NPs. The increasing porosity in NiCo+NLG-270, as compared to NiCo+NG, leads to only 25% increase of $J_{K, OER}$. Distinctly, when N-Co bonds form as in NiCo/NG or NiCo/NLG-270, $J_{K, OER}$ is dramatically promoted, suggesting a synergistic catalytic effect in the hybrids. To get rid of the SSA effect, $J_{K, OER}$ was also normalized with respect to the corresponding surface area of catalysts, and NiCo/NLG-270 exhibits three-fold enhancement in OER activity compared to NiCo/NG (Figure S22b, Supporting Information). More importantly, $J_{K, OER}$ is observed to be positively correlated to the pyridinic-N content (Figure 2f), but not the pyrrolic-N content, implying that pyridinic-N-Co bonds contribute the most to the synergistic effect for OER. With an increasing pyridinic N content in NiCo/NLG-x, $J_{K, OER}$ substantially augments up to 38 mA cm^{-2} for NiCo/NLG-270, twice that of NiCo/NG and surpassing that of RuO_2 . In addition, the electrochemical active surface area (ECSA) analysis (Figure S25, Supporting Information), a measure of the exposed active sites density,^[5b] reveals that the ECSA of NiCo/NLG-270 is more than four times as large as that of NiCo/NG, suggesting that the number of active sites is dramatically enhanced with increasing content of pyridinic-N-Co bonds. However, the physical mixtures NiCo+NLG-270 and NiCo+NG, which possess no N-Co chemical bonds, have much less active sites. At last, Nyquist plot shows that NiCo/NLG-270 has much smaller interface electron-transfer resistance than the control samples (Figure S26, Supporting Information), implying that the pyridinic-N-Co bonds afford much faster electron transfer, and in turn, promoting the OER process.

The above electrochemical measurement results strongly support that the pyridinic-N-Co bonds, rather than pyrrolic-N-Co bonds, are a principle cause of the synergistic effect which substantially enhances the reversible oxygen electrocatalysis for spinel/N-graphene. Besides superior ORR/OER activities, the hybrid (NiCo/NLG-270) also exhibits greatly improved long-term stability over the benchmark noble-metal catalysts. In

the chronoamperometric test for ORR, NiCo/NLG-270 retains 91% of the initial current density after 10 h of continuous polarization (Figure S27a, Supporting Information), whereas Pt/C catalyst shows much worse stability with only 77% current density maintained. NiCo/NG also displays worse durability than NiCo/NLG-270, suggesting that the pyridinic-N–Co bonding is also responsible to the improved catalytic stability. Meanwhile, the hybrids demonstrate much better methanol tolerance ability than Pt/C in alkaline solution (inset of Figure S27a, Supporting Information). On the other hand, the OER polarization curve of NiCo/NLG-270 recorded after 10 h testing shows negligible degradation as compared to that obtained initially (Figure S27b, Supporting Information), and the current density only degrades by less than 2 percent in the chronoamperometric response (inset of Figure S27b, Supporting Information). In contrast, RuO₂ has much worse durability in 1 M KOH solution, with a significant drop in the current density after 10 h of polarization.

To further theoretically elucidate the crucial role of pyridinic-N–Co bonds of NiCo/NLG composites in synergistic electrocatalysis for reversible ORR/OER, DFT calculations were conducted (see details in the Supporting Information). Chemical bonds between different types of nitrogen dopants (Figure S28, Supporting Information) and the surface cobalt atoms of NiCo₂O₄ as well as their influence on ORR/OER intrinsic activities were investigated. According to the theoretical model (Figure S29, Supporting Information), the surface Co are three-fold coordinated. Notably, in the case of pyridinic-N-graphene, the surface Co atom is bonded to the pyridinic N with a bond length of 1.98 Å, and the differential charge density displays certain amount of electron transfer from Co to N in the NiCo/Pyri-NG composite (Figure S30a, Supporting Information), in good agreement with the observations in the XPS and EELS measurements (Figure 1e,g,h). The moderate pyridinic-N–Co bond strength and charge transfer will remarkably facilitate activity enhancement for reversible oxygen electrocatalysis in regards to the pure NiCo₂O₄ NPs, as demonstrated explicitly below. On the contrary, there is negligible electron transfer between pyrrolic N and Co, leading to a longer Co–N bond length (2.15 Å) and very weak bonding strength for pyrrolic-N–Co (Figure S30b, Supporting Information). The Co-pyrrolic N bond is almost vertical to NiCo₂O₄ surface and the Co site is spatially confined by N doped graphene sheet, leaving no space for the adsorption of oxygen-containing intermediates. Meanwhile, the nearest Co is ≈3.15 Å away from the Co-pyrrolic N bond and is negligibly influenced by N doped graphene. Consequently, theoretically speaking, the pyrrolic-N–Co bonding could not effectively improve the ORR/OER activities, consistent with the experimental observations. Besides, in the graphitic configuration, the excess valence electrons of N are delocalized, resulting in an n-type doping effect,^[19] and making it difficult to form chemical bonds between graphitic N and Co due to electron repulsion (Figure S30c, Supporting Information). Interestingly, for the hybrid catalyst NiCo/Pyri-NG, pyridinic N, surface Co, and its neighboring Ni synergistically promote the efficiency of both ORR and OER processes. The proposed ORR/OER mechanisms on NiCo/Pyri-NG and plain NiCo₂O₄ are shown in Figure 3a and Figure S31 (Supporting Information), respectively. For Co sites on the surface of NiCo₂O₄, adsorptions of relevant species (O*, OH*, OO*,

and OOH*) tend to compensate its three-fold coordination to form Co-centered tetrahedral (O* or OH*) or pyramidal (OO* or OOH*) crystal fields (Figure 3c(i) and Figure S30, Supporting Information). As we know, NiCo₂O₄ possesses the ground-state structure of inverse spinel, suggesting that four-fold coordination is stable for Co while six-fold for Ni, as proved by previous calculations based on its lowest cohesive energy.^[16a] Consequently, for both ORR/OER over NiCo₂O₄, formation of four-fold coordinated Co is exothermic, whereas destruction of Co_{Td} crystal field is energetically unfavorable, implying that the formation of OO* (ORR) and OOH* (OER) are endothermic (Figure 3c(ii)). As expected, the calculated free energy diagram illustrates that the adsorption of O₂ (Step ①, overpotential 0.34 V) and transition from O* to OOH* (Step ②, overpotential 0.50 V) are rate-limiting for ORR and OER in an alkaline solution (Figure 3b), respectively. On the contrary, when coordinated with pyridinic N, the interfacial Co atom is four-fold coordinated and thus stabilized in NiCo/Pyri-NG (Figure 3c(ii)). Therefore, reactions which weaken (strengthen) the bond between Co (Ni) and oxygen-containing intermediates are energetically favorable, due to the fact that Co tends to occupy the tetrahedral field in regards to octahedral field for Ni. Therefore, OO* (ORR) and OOH* (OER) will coadsorb on the Co site and its adjacent Ni site in NiCo/Pyri-NG (Figure 3a,c(ii)). DFT results indicate that, due to coadsorption, the formation of OO* (ORR) and OOH* (OER), which are rate-limiting steps for plain NiCo₂O₄, become spontaneously exothermic subreactions for NiCo/Pyri-NG (Figure 3b). In this case, protonation of OO* (overpotential 0.22 V) and desorption of O₂ (overpotential 0.29 V) are the most sluggish steps for ORR and OER, respectively, and the overpotentials for reversible oxygen electrocatalysis are substantially reduced in comparison to NiCo₂O₄ alone. Furthermore, the pyridinic-N–Co bond facilitates electron storage and transfer, which acts as a sponge to storage electrons. Oxygen-containing species (O*, OH*, OO*, OOH*) will then obtain electrons from both Co and pyridinic N after being adsorbed on the active sites, thus optimizing the adsorption of the intermediates and facilitating the reversible ORR/OER processes. Additionally, the introducing of Ni also leads to a half-metallic ferrimagnetic structure of NiCo₂O₄ in which the minority spin channel becomes conducting compared to insulated Co₃O₄.^[16a,20] Thus NiCo/NLG-270 shows better ORR/OER activities than Co₃O₄/NLG-270 (Figure S32, Supporting Information).

At last, we compare the overall reversible ORR and OER electrocatalytic performance of the NiCo/NLG-270 hybrid with the benchmark noble-metal catalysts Pt/C, RuO₂, and the mixture Pt/C+RuO₂ in alkaline condition, as shown in Figure 4a. It turns out that NiCo/NLG-270 exhibits better catalytic activity than the state-of-art precious-metal-based bifunctional catalyst Pt/C+RuO₂. The overpotential between ORR and OER ($\Delta E = E_{10} - E_{1/2}$) defined as the difference between the potential required to deliver an OER current density of 10 mA cm⁻² and the ORR half-wave potential, corresponding to the efficiency loss, is usually regarded as a descriptor for evaluating the bifunctional activity of electrocatalysts.^[5b,21] An ideal reversible oxygen cathode should have an ΔE value as small as possible. Accordingly, ΔE of NiCo/NLG-270 is 0.75 V, much lower than that of Pt/C + RuO₂ (1.05 V), and also among the lowest values

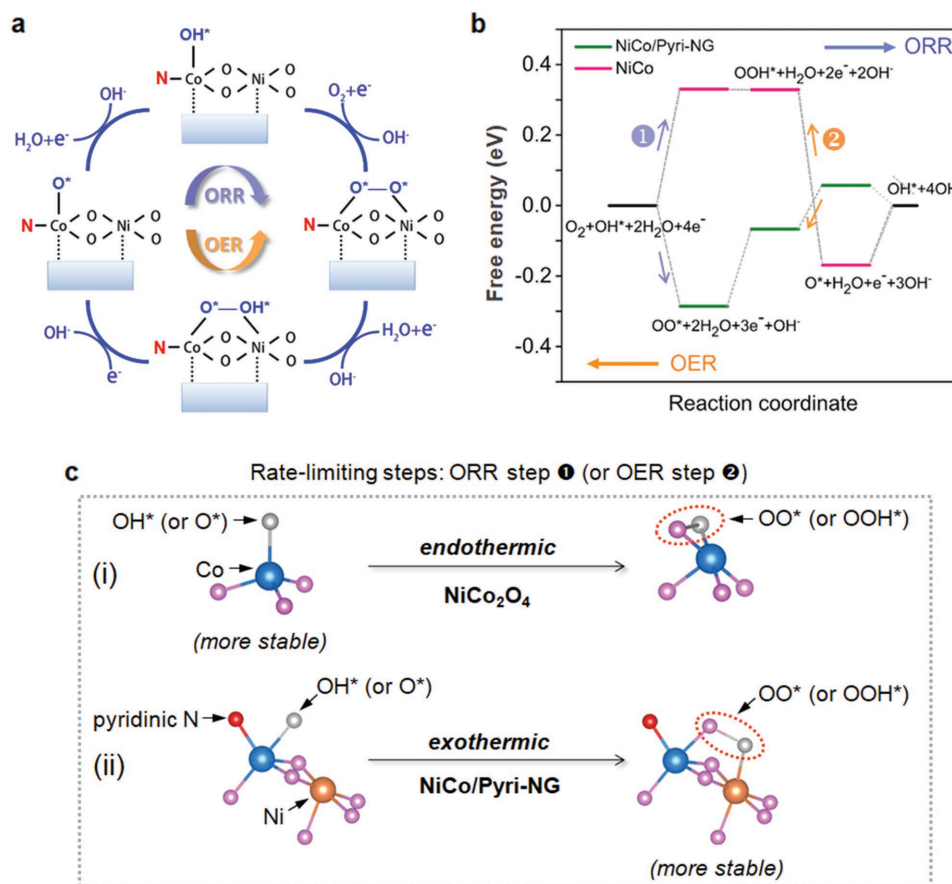


Figure 3. a) ORR (clockwise) and OER (anticlockwise) mechanisms over NiCo/Pyri-NG, where pyridinic N bonds with surface three-fold coordinated Co. b) Free energy diagram of ORR and OER processes on NiCo/Pyri-NG composite and plain NiCo₂O₄ NP. c) Adsorption configurations of oxygen-containing species on the Co site of NiCo₂O₄ (i) and NiCo/Pyri-NG (ii) for the ORR/OER rate-limiting steps. Color scheme: red for pyridinic N, blue for Co, orange for Ni, pink for O, and gray for adsorbed O^*/OH^* .

for the most active bifunctional electrocatalysts reported so far (Table S3, Supporting information). We further investigate the practical application of NiCo/NLG-270 with superior bifunctional ORR/OER capability and durability as the air cathode in a rechargeable Zn–air battery (Figure 4b). The battery can provide a high open circuit voltage of nearly 1.49 V for NiCo/NLG-270, higher than that for the mixture of commercial Pt/C and RuO₂ catalysts (Pt/C + RuO₂, 1.47 V) (Figure 4c). The galvanodynamic charge/discharge curves are shown in Figure 4d. The battery using NiCo/NLG-270 as the catalyst could offer a peak power density of 103 mW cm⁻² at 140 mA cm⁻², higher than that for Pt/C + RuO₂ (91 mW cm⁻²), besides, the rechargeable battery loaded with NiCo/NLG-270 display better discharge/charge behavior than Pt/C + RuO₂, in terms of higher discharging voltage and lower charging voltage, verifying again the superior bifunctional capability of the hybrid. The specific discharge capacity of NiCo/NLG-270 is estimated to be 403 mA h g_{Zn}⁻¹, by normalized to the mass of consumed Zn, equivalent to an energy density of 505 W h kg_{Zn}⁻¹, much larger than those of Pt/C + RuO₂ (specific capacity 310 mA h g_{Zn}⁻¹, energy density 388 W h kg_{Zn}⁻¹) (Figure S33, Supporting Information). Furthermore, the NiCo/NLG-270 catalyst allows for much lower discharge/charge overpotential (0.8 V) than the

reference catalyzed by Pt/C + RuO₂ (1.04 V) (Figure 4e). When cyclically discharge at 20 mA cm⁻² and charge at 2 mA cm⁻² for more than 40 cycles (≈15 h), only 3.8% increase of discharge/charge overpotential was observed, suggesting an excellent rechargeability for NiCo/NLG-270 catalyst, in sharp comparison with Pt/C + RuO₂ degrading after only 10 h. In terms of these results, the NiCo/NLG-270 hybrid demonstrates superior performance as reversible oxygen electrode in the Zn-air batteries relative to the noble-metal-based catalysts.

In conclusion, we developed an innovative strategy based on laser irradiation to controllably tune the relative concentrations of pyridinic and pyrrolic nitrogen dopants in the hybrid catalyst, i.e., NiCo₂O₄ NPs anchored on the N-doped mesoporous graphene nanosheets. It is found that the mesopore densities increases monotonically with higher laser beam energies, and thus creating more edges available for the formation of pyridinic N. Strong chemical bonding was verified to form between nitrogen dopants and Co ions, and we identified from experiments that, the pyridinic-N–Co bonds, instead of pyrrolic-N–Co bonds, at the hybrid interface predominantly contribute to the synergistic catalytic effect, leading to substantially enhanced ORR and OER activities. DFT calculations further revealed the crucial role of pyridinic-N–Co bonding in optimizing the

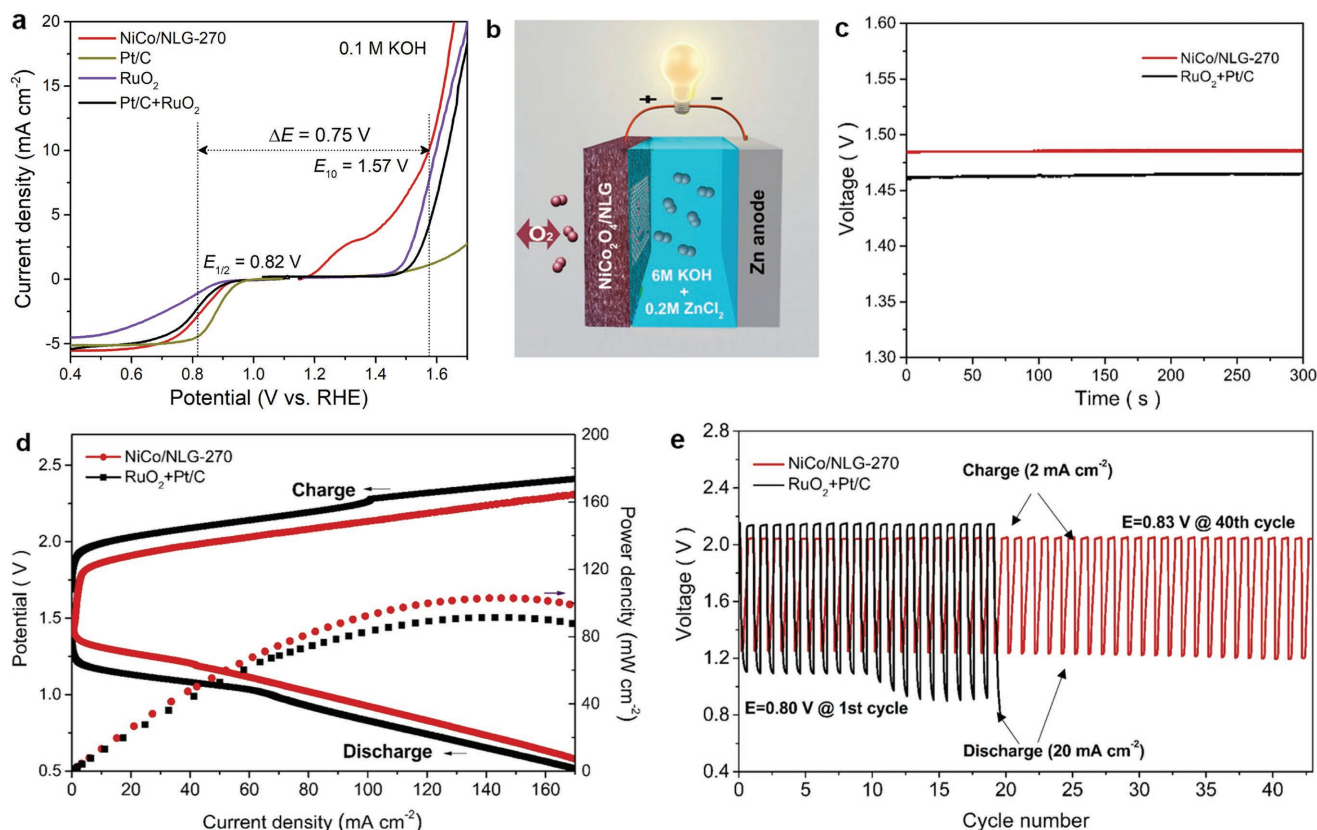


Figure 4. a) The overall polarization curves of NiCo/NLG-270, commercial Pt/C and RuO₂ catalysts, and their mixture Pt/C + RuO₂ (catalyst loading: $\approx 0.4 \text{ mg cm}^{-2}$ for all samples) in O₂-saturated 0.1 M KOH. b) Schematic illustration of the Zn–air battery. c) Open-circuit plots and d) galvanodynamic charge/discharge profiles and power density curves of NiCo/NLG-270 and Pt/C + RuO₂ catalyzed cathode. e) Discharge/recharge cycling curves at charging current density of 2 mA cm^{-2} and discharging current density of 20 mA cm^{-2} of batteries assembled with NiCo/NLG-270 and RuO₂ + Pt/C (duration: 1200 s per cycle).

adsorption of reaction intermediates and facilitating reversible ORR/OER processes. The best hybrid manifests superior bi-functional activity and stability and also excellent charge/discharge performance as the oxygen electrode for a practical Zn–air battery, outperforming the combination of benchmark noble metal catalysts Pt/C and RuO₂. This work will promote rational design of efficient hybrid electrocatalysts via interfacial structure engineering for diverse applications in renewable energy storage and conversion.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

graphenes, hybrid electrocatalysts, pyridinic nitrogen, pyrrolic nitrogen, spinels

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